

B-ES4-AIR-R2 MULTIPROTOCOL ROOM SENSOR

24 V / IDM interface and protected 5 V supply

ESP32-C6, USB-C and local sensing

0-10 V and isolated RS-485

Component List:

- J1: TEMP / 42 +6-32V / 5V
- F1: 124050H-60V 500mA
- D1: SS16 60V
- U1: LMR3610ADAR 24V to 5V
- L1: SRP1038A-330M 33uH
- D2: SS14
- U2: AP2112K-3.3TRG1
- U5: MCP4725 1280T-E/CH
- U6: OP297IDBVR
- RTH1: KTY81-210.112
- TVS1: SMBJ33A
- C1: 4.7u 100V
- C2: 100V 1u
- C3: 100V 1u
- R1: 100k 1%
- R2: 24.9k 1%
- R3: 100k
- C4: 100n 16V
- C5: 47u 10V
- C6: 47u 10V
- C7: 10u 10V
- C8: 22u 6.3V
- C11: 100n
- R9: 21k 1% CAL
- R10: 10k 1.1%
- R11: 100k 50V
- R4: 5.1k 1%
- R5: 5.1k 1%
- R6: 10k
- C9: 1u
- SW1: RESET
- SW2: BOOT
- U4: SHT45-AD5B-R2
- U7: RS485-485A
- U8: RS485-485A
- U9: RS485-485A
- U10: RS485-485A
- U11: RS485-485A
- U12: RS485-485A
- U13: RS485-485A
- U14: RS485-485A
- U15: RS485-485A
- U16: RS485-485A
- U17: RS485-485A
- U18: RS485-485A
- U19: RS485-485A
- U20: RS485-485A
- U21: RS485-485A
- U22: RS485-485A
- U23: RS485-485A
- U24: RS485-485A
- U25: RS485-485A
- U26: RS485-485A
- U27: RS485-485A
- U28: RS485-485A
- U29: RS485-485A
- U30: RS485-485A
- U31: RS485-485A
- U32: RS485-485A
- U33: RS485-485A
- U34: RS485-485A
- U35: RS485-485A
- U36: RS485-485A
- U37: RS485-485A
- U38: RS485-485A
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- U40: RS485-485A
- U41: RS485-485A
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- U95: RS485-485A
- U96: RS485-485A
- U97: RS485-485A
- U98: RS485-485A
- U99: RS485-485A
- U100: RS485-485A

Test Points:

TP1	TP2	TP3	TP4	TP5	TP6	TP7	TP8	TP9	TP10	TP11	TP12	TP13	TP14	TP15	TP16	TP17	TP18	TP19	TP20	TP21	TP22	TP23	TP24	TP25	TP26	TP27	TP28	TP29	TP30	TP31	TP32	TP33	TP34	TP35	TP36	TP37	TP38	TP39	TP40	TP41	TP42	TP43	TP44	TP45	TP46	TP47	TP48	TP49	TP50	TP51	TP52	TP53	TP54	TP55	TP56	TP57	TP58	TP59	TP60	TP61	TP62	TP63	TP64	TP65	TP66	TP67	TP68	TP69	TP70	TP71	TP72	TP73	TP74	TP75	TP76	TP77	TP78	TP79	TP80	TP81	TP82	TP83	TP84	TP85	TP86	TP87	TP88	TP89	TP90	TP91	TP92	TP93	TP94	TP95	TP96	TP97	TP98	TP99	TP100
24V_RAW	24V_PROT	SYS_5V	+3V3	I2C_SDA	I2C_SCL	RH_OUT	TEMP_KTY	RS485_A	RS485_B	RS485_C	RS485_D	RS485_E	RS485_F	RS485_G	RS485_H	RS485_I	RS485_J	RS485_K	RS485_L	RS485_M	RS485_N	RS485_O	RS485_P	RS485_Q	RS485_R	RS485_S	RS485_T	RS485_U	RS485_V	RS485_W	RS485_X	RS485_Y	RS485_Z	RS485_AA	RS485_AB	RS485_AC	RS485_AD	RS485_AE	RS485_AF	RS485_AG	RS485_AH	RS485_AI	RS485_AJ	RS485_AK	RS485_AL	RS485_AM	RS485_AN	RS485_AO	RS485_AP	RS485_AQ	RS485_AR	RS485_AS	RS485_AT																																														

Field I/O: push-in terminals, protected 1-Wire, dry contacts and 0-10 V input
ENGINEERING SAMPLE – BENCH VALIDATION REQUIRED BEFORE IDM CONNECTION

ENGINEERING SAMPLE – BENCH VALIDATION REQUIRED BEFORE IDM CONNECTION

Sheet: / ~~MX205R~~ MX205R PUSH-IN / 3V3 / 1-WIRE ~~MX205R~~ PUSH-IN
File: IDM-RoomSensor-ESP-B-ES4-AIR-R2.kicad sch

File: IDM-RoomSensor-ESP-B-ES4-AIR-R2.kicad_sch

Title: IDM Room Sensor - B-ES4-AIR-R2 / US / RS-485 / 0-21V

Size: A4 Date: 2026-08-30 Rev: B-ES4-AIR-ES4

UT	KiCad E.D.A. 10.0.5	TEMP_KTY	RS485_A	RS485_B	RS485_COM	GND
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4	R14	R15	5	TVS2	R16	6	R17
100R	4.7k	PCESD3V3D3	1k	1k	1k	1k	1k

TEMP_1 P_n1 RS485_A P_n1 RS485_B P_n1 RS485_COM P_n1 GND P_n1